

Estela Pérez Luque, PhD

Human Factors Engineer

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PROFILE

Human Factors Engineer with 6+ years of experience planning, executing, and analysing evaluation and validation studies in complex engineering environments. My background combines HF modelling, testing, simulation, digital human modelling, motion capture, and structured data analysis to assess posture, visibility, reachability, usability, and operator interaction, and to generate traceable evidence for design and engineering decisions. I have worked with Volvo Cars, Volvo Trucks, and Scania, translating user behaviour, technical constraints, and evaluation results into requirements, test activities, and actionable recommendations.

TECHNICAL SKILLS

Core tools: MATLAB, Python, Excel

Testing & Validation: V&V, HF test planning, protocol design, motion capture, multi-sensor acquisition, time-series analysis, technical reporting, traceability

Human Factors: HMI, usability, task analysis, workload-related assessment, physical and cognitive ergonomics, anthropometric variability, posture, visibility, reachability

Simulation & CAD: IPS IMMA, Siemens Jack, AnyBody, Autodesk Inventor, SolidWorks

Languages: English (C1), Spanish (Native), Swedish (B1)

PROFESSIONAL EXPERIENCE

Human Factors Researcher - University of Skövde, Sweden (Onsite)

Sep 2020 – Oct 2025

- **HF test planning and execution:** Planned and executed Human Factors studies assessing posture, visibility, reachability, comfort, and task execution across multiple studies and operating conditions; managed participant selection, protocols, multi-sensor acquisition, and reporting for datasets of up to 199 participants.
- **Evaluation methodology and analysis:** Applied quantitative and qualitative methods to compare predicted and observed operator behaviour, including posture and movement patterns, identifying 15 to 20% variation depending on evaluation approach and showing its impact on conclusions and recommendations.
- **Simulation-based testing and validation:** Used IPS IMMA and motion capture to compare simulated and real-world behaviour across multiple vehicle configurations, supporting more robust early assessment of visibility, reachability, seating posture, and control access.
- **Anthropometric population sampling:** Defined representative test populations using percentile-based sampling, anthropometric variability frameworks, and population ellipsoids, improving the robustness of accommodation and ergonomics assessments across target user groups.
- **V&V documentation and stakeholder coordination:** Structured evaluation criteria, protocols, and technical reports with clear traceability of requirements and decisions, while coordinating HF input with Volvo Cars, Volvo Trucks, and Scania.
- **Knowledge transfer and instruction:** Communicated results through 20+ publications, 15+ international presentations, and teaching in CAD and digital human modelling for Bachelor's and Master's students, supporting uptake of simulation-based Human Factors methods.

Visiting Researcher - University of Antwerp, Belgium (Onsite)

Sep 2024 – Mar 2025

- **External campaign coordination:** Led the setup of an experimental data collection campaign in an external lab, including ethics approval, protocol design, recruitment, and cross-institution coordination, ensuring test readiness and protocol consistency.

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- **Multi-sensor acquisition and processing:** Designed motion capture and 4D body scanning protocols for 30+ participants and built reproducible processing pipelines, improving repeatability, traceability, and cross-study comparability.
- **Documentation and compliance:** Contributed to ethics and data governance documentation, reinforcing traceability and compliance-minded collaboration across institutions.

Co-founder and R&D Lead - Swe-Ex AB, Sweden (Startup)

Sep 2021 – Mar 2023

- **Requirements definition and prototype testing:** Translated user needs and safety requirements into measurable HF requirements and evaluation criteria, guiding three prototype iterations from CAD concept to physical validation.
- **Evidence-based design iteration:** Combined user feedback with technical assessment to evaluate fit, usability, and ergonomic performance, enabling faster iteration and better decisions under real project constraints.

Research Assistant - University of Skövde, Sweden (Onsite)

Jun 2019 – Sep 2020

- **Applied HF studies with industry partners:** Developed and executed simulation studies across 6+ industry projects, analysing posture, reachability, visibility, and ergonomics and producing reports, demonstrations, and recommendations to support design decisions.
- **DHM training and instruction:** Created tutorials and training materials on digital human modelling tools and taught CAD and ergonomics courses, supporting adoption of simulation-based HF methods in academic and applied engineering settings.

EDUCATION

PhD in Informatics, University of Skövde (Sweden)

Sep 2020 – Oct 2025

[PhD Thesis](#): Human Posture and Motion Prediction for Automotive Ergonomics Design

Focus: digital human modelling, simulation-based evaluation, V&V, anthropometric variability, ergonomics for complex product development

Master of Science in Intelligent Automation, University of Skövde (Sweden)

Aug 2018 – Jun 2019

Bachelor of Science in Mechanical Engineering, University of Skövde (Sweden)

Aug 2017 – Jun 2018

Degree in Mechanical Engineering, University of Málaga (Spain)

Aug 2012 – Jun 2018

LEADERSHIP & RECOGNITION

- Elected Student Representative and Board Member in doctoral committees, representing 40+ PhD students in university governance and program improvement decisions; and recipient of multiple international grants and awards, including travel and conference funding supporting research dissemination and cross-institutional collaboration.

RESEARCH PUBLICATIONS

Full publication list available at: [Google Scholar – Estela Pérez Luque](#)